

BIODIVERSITY

Biodiversity is the variation among living organisms from different sources including terrestrial, marine and desert ecosystems, and the ecological complexes of which they are a part."

IMPORTANCE

- ▶ **Biodiversity is essential for the processes that support all life on Earth, including humans. Without a wide range of animals, plants and microorganisms, we cannot have the healthy ecosystems that we rely on to provide us with the air we breathe and the food we eat. And people also value nature of itself.**
- ▶ Pollinators such as birds, bees and other insects are estimated to be responsible for a third of the world's crop production.
- ▶ Soil is teeming with microbes that are vital for liberating nutrients that plants need to grow
- ▶ Trees, bushes and wetlands and wild grasslands naturally slow down water and help soil to absorb rainfall. When they are removed it can increase flooding.
- ▶ Many of our medicines, along with other complex chemicals that we use in our daily lives such as latex and rubber, also originate from plants. Spending time in nature is increasingly understood to lead to improvements in people's

- Scientists have estimated that there are around 8.7 million species of plants and animals in existence. However, only around 1.2 million species have been identified and described so far, most of which are insects.

Types of Biodiversity

There are the following three different types of biodiversity:

- Genetic Biodiversity
- Species Biodiversity
- Ecological Biodiversity

Genetic Diversity	Species Diversity	Ecological diversity
Population	Kingdom	Biomes
Individual	Phyla	Bio regions
Chromosomes	Family	Ecosystem
Genes	Genera	Landscape
Nucleotides	Species	Habitats
	Subspecies	Niches
	Population Individual	Population



There are three types of biodiversity

➤ Diversity of Species



➤ Diversity of Ecosystem



➤ Diversity of Genes



Measuring Diversity And The threats to Biodiversity

- ▶ Two components are used
 - The number of entities(species)
 - The degree of dissimilarities between those entities

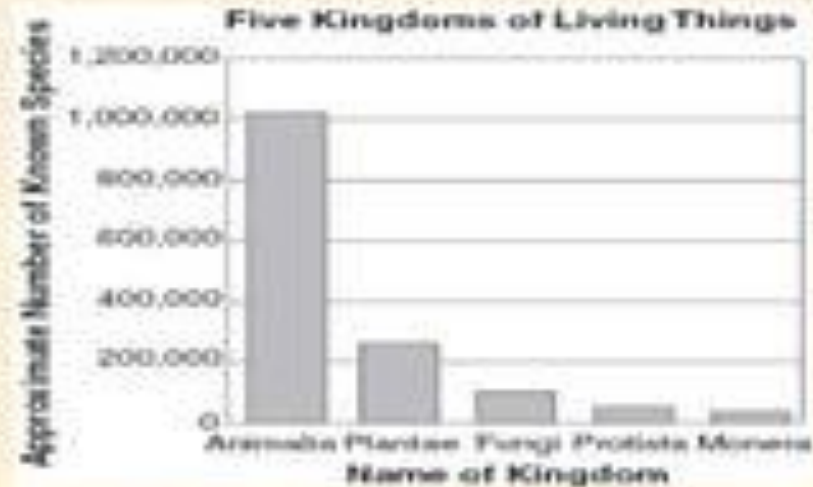
The growing human population and the land development that comes with population growth can be especially detrimental because land conversion and subsequent loss of habitats can affect the stability or continued existence of species.

DISTRIBUTION OF BIODIVERSITY

- ❖ Flora and fauna diversity depends on-
 - Climate
 - Altitude
 - Soils
 - Presence of other species
- ❖ Most of the biodiversity concentrated in Tropical region.

❖ BIODIVERSITY HOTSPOTS:

- A region with high biodiversity with most of species being Endemic.
- India have two Biodiversity Hotspots- East Himalayan Region and Western Ghat.



HABITAT LOSS:

FACTORS INFLUENCING HABITAT LOSS:

- Deforestation
- Destruction of wetlands
- Habitat fragmentation
- Raw materials
- Production of drugs
- Illegal trade
- Developmental activities

- With the current rate of loss of forest habitat, it is estimated that 20 – 25% of the global flora and fauna would be lost within few years.

POACHING:

1. **Subsistence poaching** – killing animal to provide enough **food for survival**
2. **Commercial poaching** – hunting and killing animals to **sell their products**

Factors influencing Poaching:

- ☐ Human population
- ☐ Commercial activities

Remedial Measures:

- ☐ Illegal hunting and trading should be stopped
- ☐ Not purchase fur coat, purse or bag made of crocodile or python skin
- ☐ Bio – diversity laws should be strengthened

- Habitat loss is a challenge for virtually all species, as humans convert natural habitats to other land uses.
- Overexploitation from extractive uses, such as commercial fishing and game hunting, can greatly reduce species numbers, sometimes to the brink of extinction¹.
- ▶ The Food and Agriculture Organization (FAO) estimates that in 2008, approximately 32% of fish stocks were overexploited, depleted, or recovering from depletion².
- ▶ Upsetting the viability of a single species can have far-reaching impacts for the balance of an entire ecosystem.
- ▶ People can also harm biodiversity by putting stress on environments and species through overuse. Outdoor recreation may be a positive, healthful activity for humans, but high numbers of visitors to an area can damage plant life, stress local animal populations, and introduce invasive species.
- ▶ Invasive species can outcompete or consume native species to the point of extinction.
- ▶ All forms of pollution, from chemicals to nutrient loading, can also pose serious threats to aquatic and terrestrial species.

RECENT ISSUES ON BIODIVERSITY

- Some 75 per cent of the genetic diversity of crop plants been lost in the past century.
- Some scientists estimate that as many as 3 species per hour are going extinct and 20,000 extinctions occur each year.
- Roughly one-third of the world's coral reef systems have been destroyed or highly degraded.
- About 24 per cent of mammals and 12 per cent of bird species are currently considered to be globally threatened.
- More than 50 per cent of the world's wetlands have been drained, and populations of inland water and wetland species have declined by 50 per cent between 1970 and 1999.

ENDANGERED AND ENDEMIC SPECIES OF INDIA:

- According to International Union Conservation of Nature and Natural Resources – IUCN the species are classified as follows,
 - **Extinct Species** – No longer found
 - **Endangered Species** – Number of species has been reduced to critical level
 - **Vulnerable Species** – Continuous decline due to habitat destruction
 - **Rare Species** – Species located within a restricted area or thinly scattered over a more extensive area.

► The benefits of conserving biodiversity

- Biodiversity supports food security and sustained livelihoods through overall genetic diversity.
 - Genes regulate all biological processes on the planet and increase the ability of organisms to cope with environmental stressors.
 - Preserving genetic diversity ensures the continuing existence of a wide-range of crops that may be able to withstand disease, and potentially useful biochemicals such as those used in healthcare. It also means availability of species for pollination and pest control. Losses in genetic diversity will decrease organisms' coping ability and risk losing potentially beneficial biological information³.
 - Biodiversity has greatly contributed to modern medicine and advancements in human health research and treatment. Many modern pharmaceuticals are derived from plant species, including the anti-tumor agent Taxol from the Pacific yew tree, the anti-malarial artemisinin from sweet wormwood, and the cardiac drug digoxin from the digitalis plant.
 - Pharmaceuticals can also be derived from non-plant species, such as the drug ziconotide, which has been highly effective in relieving nerve pain and severe pain in cancer patients and is derived from the venom of predatory cone snails⁴.

- Without the species that provide these drugs, it is possible that treatments for ailments like malaria, tuberculosis, cancerous tumors, congestive heart failure and multiple other illnesses may never have been discovered.
- As conversion of habitats and subsequent losses in diversity take place, the potential for losing cures for some of the world's most troubling ailments increases.
- In addition to the many medicinal benefits from biodiversity, human health can be positively affected simply by spending time in outdoor environments, which has been linked to increases in life satisfaction and happiness, and decreases in blood pressure, anxiety, and cardiovascular disease symptoms.
- Conserving biodiversity and protecting a wide range of habitats maintains the many benefits that this diversity provides for all species. Highly diverse environments, are prime ecosystems that support many species in addition to being aesthetically beautiful, educational, and interesting recreation sites.
- Biodiversity conservation efforts are essential in maintaining functioning ecosystems, a steady food supply, and the multiple other benefits including aesthetics, recreation, and spiritual purposes to tribal nations

Ecological services:

- Balance of nature
- Biological productivity
- Regulation of climate
- Degradation of waste
- Cleaning of air and water
- Cycling of nutrients
- Control of potential pest and disease causing species
- Detoxification of soil and sediments
- Stabilization of land against erosion
- Carbon sequestration and global climate change
- Maintenance of Soil fertility

► **Types of Conservation**

- Conservation of biodiversity can be done in two different types, namely the ex-situ conservation, in-situ conservation
- Ex situ conservation: Ex situ conservation means conserving the areas beyond where they naturally occur. Animals and plants that are reared or cultivated in areas like zoological or botanical parks come under this method of conservation. Introducing an animal or plant which has become extinct back to its habitat is also a form of ex situ conservation. Seed Banks, botanical, horticultural and recreational gardens are important centres for ex situ conservation.
- In situ conservation: In situ conservation means conserving the areas within the natural habitat of animals and plants. The establishment of natural reserves, national parks, sanctuaries, etc. comes under in situ conservation. It is a method which is cost-effective and convenient to conserve biodiversity. Organisms live a natural ecosystem and thereby evolve more efficiently and adjust to various environmental conditions. Also, a large number of living organisms can be conserved under a natural habitat.

TYPES OF BIODIVERSITY CONSERVATION:

- **In – situ Conservation** : Involves **protection of fauna and flora within its natural habitat.**

In – situ Conservation Methods	Numbers available in india
Biosphere Reserves	18
National Parks	104
Wild – life Sanctuaries	543
Botanical Gardens	30



MERITS OF IN – SITU CONSERVATION:

- Very cheap and convenient method
- Adjusted to natural disasters like drought, floods, forest fires, etc.

LIMITATIONS OF IN – SITU CONSERVATION:

- Large surface area is required for preservation
- Maintenance is not proper due to shortage of staff and pollution

METHODS OF EX – SITU CONSERVATION:

- **National Bureau of Plant Genetic Resources (NBPGR)** – preserves agricultural and horticultural crops
- **National Bureau of Animal Genetic Resources (NBAGR)** – preserves the semen of domesticated bovine animals
- **National Facility for Plant Tissue Culture Repository (NFPTCR)** – conservation of varieties of crop plants or tree by tissue culture

MERITS OF EX – SITU CONSERVATION:

- Survival increase due to special care and attention
- Longer life span due to assured food, water, shelter and security
- It is carried in case of endangered species

LIMITATIONS OF EX – SITU CONSERVATION:

- Expensive method
- Freedom of wildlife is lost
- Animal cannot survive in natural environment
- Adopted only for few selected species

- ▶ **WHAT IS A BIODIVERSITY HOTSPOT?**
- ▶ There are currently **36 recognized biodiversity hotspots**. These are Earth's most biologically rich—yet threatened—terrestrial regions.
- ▶ To qualify as a biodiversity hotspot, an area must meet two strict criteria:
 - Contain at least 1,500 species of vascular plants found nowhere else on Earth (known as "endemic" species).
 - Have lost at least 70 percent of its primary native vegetation.

HOT SPOTS OF BIODIVERSITY:

- Areas which exhibit **high species richness** as well as high species endemism are termed as **Hot spots of Biodiversity**.
- There are **36 hot spots of biodiversity** on a global level and *four* are present **in India i.e. Himalayas, Western Ghats, Indo-Burma, & Sundaland**.
- At global level, these are the areas of **high conservation** priority, if these species lost, they can never be **replaced or regenerated**.
- Hotspot covering less than 2% of world's land are found to have 50% of terrestrial biodiversity.
- About 40% of terrestrial plants and 25% of vertebrate species are endemic and found in hotspot.

BIODIVERSITY CONVENTIONS

- The first convention on biodiversity organized at Rio De Janeiro, capital of Brazil from June 5 to 16, 1992 named as United Nation Conference On Environment and Development (UNCED), better known as Rio Summit to maintain ecological balance and enrich biodiversity. The agreement on biodiversity signed by 150 countries including three programmes-
 - ❖ To ensure conservation of biodiversity
 - ❖ Sustainable use of biodiversity
 - ❖ Rational and equitable share of profit to accrue from use of genetic resources.
- The second convention organized at Johannesburg in 2002 called World Summit On Sustainable Development (WSSD) where the Biodiversity and Sustainable Ecosystem Management was the issue.